

Accelerate Boltzmann Sampling of Multimodal Distributions by Training with Log-Likelihood Ratio Discrepancy

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1 Introduction

The Boltzmann generator based on normalizing flows (Noé et al., 2019; Wu et al., 2020) has been a crucial approach to sampling complicated distributions, but on multimodal targets normalizing flows trained with the reverse KL are prone to mode collapse, where the modes lost during training corrupt the sample quality. To address this issue, several approaches have been proposed to mitigate mode collapse during training.

Suppose μ is the target distribution on $\mathbb{R}^d$ and ν is the pushforward distribution of the trained flow. The first approach is to replace the mode-seeking reverse KL by the mass-covering forward KL

$$\text{KL}(\mu\|\nu) = \int_{\mathbb{R}^d} \mu(x) \log \frac{\mu(x)}{\nu(x)} dx, \quad (1)$$

which is more stable on multimodal targets. Although the forward KL inherently requires samples from the target distribution, approximate target samples can be efficiently generated by Annealed Importance Sampling (AIS), which uses a sequence of reweighting and resampling steps to transport the pushforward distribution $\nu(x)$ to the target distribution $\mu(x)$.

Another representative method is the Flow AIS Bootstrap (FAB) (Midgley et al., 2022), which uses the α -divergence with $\alpha = 2$,

$$D_2(\mu\|\nu) = \int_{\mathbb{R}^d} \frac{\mu^2(x)}{\nu(x)} dx, \quad (2)$$

which remains purely energy-based and is capable of discovering new modes. Since the forward KL can be viewed as the limit of the α -divergence as $\alpha \rightarrow 1$, FAB can be seen as a generalization of the forward KL to a different choice of α . Compared to the forward KL, FAB requires importance sampling against the more delicate distribution μ^2/ν , which may cause training stability issues.

Although these approaches are theoretically capable of mitigating mode collapse, their training speed and stability are still limited by the complexity of the potential landscape. In many cases we already know the secondary modes of the target distribution μ , for example the multiple configurations of a molecule; can we integrate these known modes into the flow training?

In this paper we propose a new functional, the log-likelihood ratio discrepancy, defined by

$$\text{XX}(\mu\|\nu) = \iint_{\mathbb{R}^d \times \mathbb{R}^d} \mu(x)\mu(y) \left| \log \frac{\mu(x)}{\nu(x)} - \log \frac{\mu(y)}{\nu(y)} \right| dx dy. \quad (3)$$

The shorthand XX denotes this nonnegative functional and is used purely for notational convenience. The XX operates in a fundamentally different regime from the forward KL and FAB: while the forward KL and FAB are based on a global Jensen’s inequality, the XX works locally and drives the variation of the log-likelihood ratio $\log(\mu/\nu)$ to zero. This means that inaccurate importance-sampled estimates of the target $\mu(x)$ do not shift the minimizer, so the XX can be combined with known target data. Overall, the XX is not constructed to replace the forward KL or FAB, but is designed for specific use cases:

- It serves as a local complementary loss and can be freely combined with any global loss such as the forward KL or FAB;
- It accelerates flow training and strengthens mode discovery;
- It exploits known target-mode information, yielding a mixed energy-data approach.

Moreover, the XX has the same scale as the usual KL divergence (its integrand is first-order in the log-likelihood ratio), so no additional tuning is generally required.

2 Single-step training loss

Let $U_0(x)$ and $U_1(x)$ be the source and target potential functions on $\mathbb{R}^d$, and let $\mu_0 \propto \exp(-U_0)$ and $\mu_1 \propto \exp(-U_1)$ be the corresponding source and target distributions. Given sufficient samples from μ_0 , we aim to construct a differentiable flow map G on $\mathbb{R}^d$ such that $G_{\#}\mu_1 \approx \mu_0$, so that target samples can be obtained from $G_{\#}^{-1}\mu_0$ via importance sampling.

We first review the construction of the training loss for G under the forward KL and FAB, then derive the XX functional for training G .

2.1 Forward KL

For the KL divergence, we have the symmetry equality $\text{KL}(G_{\#}\mu_1\|\mu_0) = \text{KL}(\mu_1\|G_{\#}^{-1}\mu_0)$. The pushforward density of $G_{\#}^{-1}\mu_0$ is given by

$$(G_{\#}^{-1}\mu_0)(y) = \mu_0(G(y))|\det J_G(y)|, \quad (4)$$

hence the forward KL reads

$$\begin{aligned} \text{KL}(G_{\#}\mu_1\|\mu_0) &= - \int_{\mathbb{R}^d} \mu_1(y) \log(G_{\#}^{-1}\mu_0)(y) dy + \text{const} \\ &= \int_{\mathbb{R}^d} \mu_1(y) \left(U_0(G(y)) - \log |\det J_G(y)| \right) dy + \text{const}. \end{aligned}$$

The forward KL training loss in terms of G is defined by

$$\mathcal{L}_{\text{KL}}[G] = \mathbb{E}_{y \sim \mu_1} \left[U_0(G(y)) - \log |\det J_G(y)| \right]. \quad (5)$$

Here, we deliberately use the inverse map G from target to source so that $\mathcal{L}_{\text{KL}}[G]$ does not involve the inverse of G , for the best efficiency of training.

The forward KL requires explicit samples from the target distribution μ_1 , which can be approximately obtained via the Annealed Importance Sampling (AIS) approach. To be specific, we connect the pushforward distribution $G_{\#}^{-1}\mu_0$ and the target μ_1 with the following $M+1$ intermediate distributions:

$$\pi_k = (G_{\#}^{-1}\mu_0)^{1-\frac{k}{M}} (\mu_1)^{\frac{k}{M}}, \quad k = 0, 1, \dots, M. \quad (6)$$

Equivalently, their negative log-likelihood is explicitly constructed as

$$\begin{aligned} -\log \pi_k(y) &= -\left(1 - \frac{k}{M}\right) \log(G_{\#}^{-1}\mu_0)(y) - \frac{k}{M} \log \mu_1(y) \\ &= \left(1 - \frac{k}{M}\right) \left(U_0(G(y)) - \log |\det J_G(y)| \right) + \frac{k}{M} U_1(y) + \text{const}. \end{aligned} \quad (7)$$

The AIS is then constructed as:

Algorithm 1: Annealed Importance Sampling (AIS) from $G_{\#}^{-1}\mu_0$ to μ_1

Input: source and target potentials U_0 and U_1 , flow map G

We approximate μ_1 samples by importance sampling against the trained flow: - Draw $x_1, \dots, x_N \sim \mu_0$ (source); - Push forward to $y_i = G^{-1}(x_i)$, so $y_i \sim G_{\#}^{-1} \mu_0$ (detach to stop gradient); - Compute the log-weights $\log w_i = U_0(x_i) - U_1(y_i) - \log |\det J_G(y_i)|$; - Resample $\{y_i\}$ by weights w_i to obtain approximate samples from μ_1 ; - Apply a few Langevin steps on $\{y_i\}$ to obtain approximate samples of μ_1 .

Because of the resampling, the approximation to μ_1 is biased, so a large batch size is recommended; the parameter λ then becomes a trade-off between overcoming mode collapse and sampling accuracy.

References

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